

# Canonical Universal Translation Pipeline

( \$ . . )

This directory ships the **inherited, mandatory** translation standard for every project that inherits the HelixConstitution. Per **Constitution.md §11.4.133**, ALL translations of ANY content (UI strings, documents, articles, CVs, READMEs, marketing copy, in-app messages, release notes, legal text — any human-language string rendered in a non-source language) MUST be produced by **HelixTranslate** ( `git@github.com:HelixDevelopment/HelixTranslate.git` ) driven through the scripts in this directory. This is a hard, non-negotiable constraint — not a recommendation.

## Purpose

A single, evidenced, fail-loud translation engine + pipeline so that every translated string in the fleet has a known provenance, a reproducible pipeline, and a captured accuracy proof. One engine, one pipeline, one evidence trail.

## Files

| Script                             | Purpose                                                                                                                                                                                            |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>translate-pipeline.sh</code> | Universal Markdown / plain-text translator. Two modes: <b>plain</b> (whole file) and <code>--article</code> (YAML frontmatter kept byte-identical; only the body is translated, then reassembled). |
| <code>render-articles.sh</code>    | Renders translated article Markdown sources into standalone HTML fragments (frontmatter <code>title / repo / tech</code> → header, body → <code>pandoc html5</code> ).                             |

Both scripts are **reusable verbatim** across projects and languages. Do not fork or hand-edit per project; configure them via environment variables (below).

## Usage

```
# Translate a whole Markdown/plain file:
translate-pipeline.sh --in <src.md> --out <dst.md> --lang <ru|sr>

# Translate an article (preserve YAML frontmatter byte-identical, translate body):
translate-pipeline.sh --in <src.md> --out <dst.md> --lang <ru|sr> --article

# Render translated article sources to HTML fragments:
# SOURCES: <site-root>/_article_src/<lang>/*.md
# OUTPUT: <site-root>/articles/<lang>/*.html
render-articles.sh <site-root> <lang> [<lang> ...]
```

Language → script mapping: `ru` → `cyrillic`, `sr` → `latin`.

Exit status: `0` only on a real, engine-produced translation. On failure the destination is **NOT written** and the script exits non-zero so callers detect it.

## Provider configuration

Routing is **explicit and recorded** — never implicit/defaulted in the dark.

| Variable                            | Default                                                                    | Meaning                                                                                                                                 |
|-------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| <code>HELIX_TRANSLATE_BIN</code>    | <code>/Volumes/T7/Projects/helix_translate/build/unified-translator</code> | Path to the HelixTranslate <code>unified-translator</code> engine binary (built from the HelixTranslate repo by the consuming project). |
| <code>TRANSLATE_EVIDENCE_DIR</code> | <code>&lt;project&gt;/_tests/evidence/translate</code>                     | Where per-file run logs (evidence) are appended.                                                                                        |

Canonical provider routing (in-script):

- **Primary:** provider `groq` , model `llama-3.3-70b-versatile`
- **Fallback:** provider `mistral` , model `mistral-large-latest`
- On rate-limit / error: retry up to 3× with linear backoff, then fail over to the fallback provider. The provider that actually produced the output is recorded in the per-file log.

Any deviation from this canonical routing **MUST** be stated explicitly in configuration and logged.

## Hard rules (§ . . )

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- **No hand-translation.** The engine is the single source of every translated string.
- **No other engines.** No Google Translate, DeepL, raw ChatGPT/Claude prompting, OS translation APIs, or library i18n auto-translators. Exactly one engine: HelixTranslate.
- **No silent fallback.** No quiet substitution of the source string, no empty/partial output passed off as a translation, no swallowed engine error, no "absence-of-error" PASS. On exhaustion of all providers the pipeline **FAILS LOUD** (non-zero exit, destination not written). Provider-to-provider failover **within** HelixTranslate (primary → fallback, with logged retries) is permitted and is **NOT** a silent fallback; falling back to any non-HelixTranslate source is forbidden.
- **Mandatory validation.** Every generated translation **MUST** be validated for accuracy with real (physical) captured evidence (per §11.4.5 / §11.4.69 / §11.4.107). The per-file run log (provider used, byte counts, retries, success/failure) is the evidence artifact. An unvalidated or unevidenced translation is treated as ABSENT.

## Inherited standard

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This is the **inherited** standard. Consuming projects **MUST** restate + cite §11.4.133 via §11.4.35 inheritance, supply their own source files / target languages, and build the HelixTranslate engine from `git@github.com:HelixDevelopment/`

`HelixTranslate.git` . Non-compliance is a release blocker. No escape hatch — no `--allow-hand-translation` , `--other-engine-OK` , `--skip-translation-validation` , or `--silent-fallback` flag exists.

See also: `docs/translations.md` and `Constitution.md §11.4.133`.